



IT Director/manager/ supervisor Guideline - Pre-Deployment Checklist and Gates version 1/2020

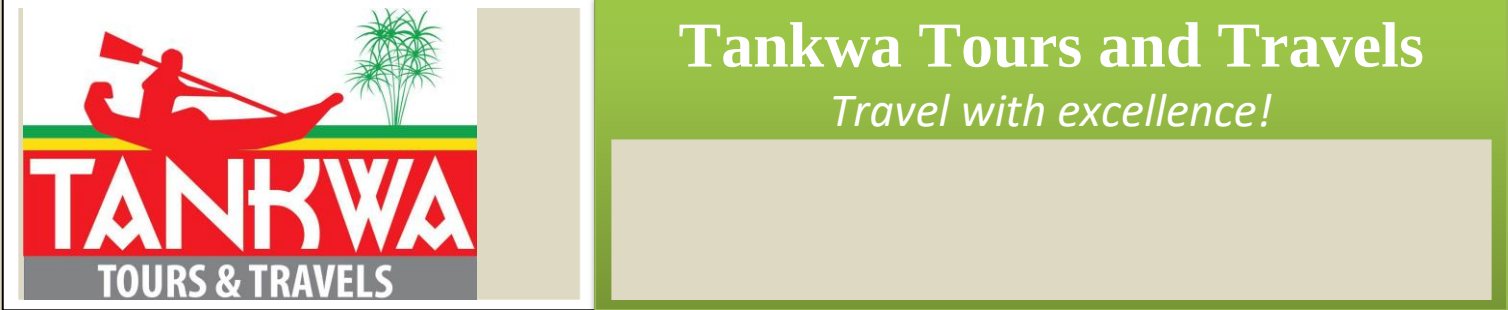
Before deployment can even be scheduled, several mandatory checks (often called "**gates**") must be passed.

- **Code Review and Merging:** The new code must be fully reviewed by peers (other developers/engineers) and merged into the main release branch.
- **Successful Builds:** The code must compile and build successfully on the Continuous Integration (CI) server.
- **Unit and Integration Testing:** All automated **Unit Tests** (testing individual components) and **Integration Tests** (testing how components work together) must pass with a predefined success rate (e.g., 90% or 100%).
- **Security Scanning:** Static and dynamic code analysis tools must be run to check for common security vulnerabilities (e.g., SQL injection, XSS). Critical findings must be resolved.
- **Documentation and Runbook:**
 - **User Documentation:** Any new features or changes need updated user guides or release notes.
 - **Deployment Runbook:** A step-by-step, documented plan for the deployment and rollback process must be created and often reviewed by operations or site reliability teams.

Staging and Quality Assurance (QA) Testing

This is the most crucial phase where the code is tested in an environment that **closely mirrors the production site**.

- **Environment Setup:** The code is deployed to the **Staging** or **Pre-Production** environment. This environment should have the same hardware, operating system, database version, and configuration as the main site.
- **System and User Acceptance Testing (UAT):**
 - **Functional Testing:** QA testers verify that all new features work as intended and that existing features (**regression testing**) haven't been broken.
 - **UAT:** Business users or product owners test the features to ensure they meet the original requirements and are usable from a business perspective.
- **Performance and Load Testing:**



- Simulated user traffic is sent to the staging environment to ensure the new code can handle the expected load (and often more) without performance degradation or crashes. Key metrics like response time are monitored.
- **Disaster Recovery/Rollback Testing:** The team *simulates* a failure during deployment or immediately after to ensure the established **rollback plan** (how to revert to the old working version) is effective and quick.

Final Preparation and Communication

Once testing is complete and successful, the final logistical steps are taken.

- **Change Management Approval:** A formal request for change (RFC) is submitted to a **Change Advisory Board (CAB)** or equivalent body. This review checks for potential risks, impact on other systems, and ensures the deployment window avoids peak traffic times.
- **Notifications:** All relevant stakeholders are informed about the upcoming deployment, including:
 - Operations/Support teams (for monitoring).
 - Business teams (for readiness and communication to customers, if needed).
 - The expected **maintenance window** or downtime (if any).
- **Final Backups:** **Full backups** of the production database and application configurations are taken immediately before the deployment begins.

The A-Z IT Deployment Pipeline

Phase	Steps (A-Z Flow)	Key Goals
I. Develop & Commit (CI Start)	A. Code Changes: Developer writes and tests code locally, ensuring it works in their isolated environment.	Feature Completion
	B. Commit & Push: Code is committed to a Version Control System (VCS) like Git (e.g., GitHub, GitLab) and pushed to a shared branch.	Code Integration



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Phase	Steps (A-Z Flow)	Key Goals
	(Trigger)	
II. Continuous Integration (CI)	C. Build Trigger: The code push automatically triggers the CI/CD pipeline server (e.g., Jenkins, GitLab CI).	Automation Start
	D. Code Compilation: The source code is fetched and compiled into executable binaries, libraries, or container images (artifacts).	Working Artifact
	E. Unit Testing: Automated unit tests run against the newly built code to check the smallest components for correctness. (Must Pass).	Component Quality
	F. Static Analysis: Tools scan the code for style violations, complexity, and basic security flaws (e.g., Linters, SAST).	Security & Cleanliness
	G. Artifact Storage: The successfully built and tested artifact is stored in a secure repository (e.g., Artifactory, Docker Registry) with a unique version tag.	Single Source of Truth
III. Continuous Testing (CD Start)	H. Deployment to DEV/TEST: The artifact is automatically deployed to a non-production (Development/Test) environment.	Environment Verification
	I. Integration Testing: Tests run to ensure the new component works correctly with other systems, APIs, and databases.	System Cohesion
	J. Staging Deployment: The same artifact is promoted and deployed to the Staging/Pre-Production environment, which is an exact replica of the main site.	Production Mirror
	K. User Acceptance Testing (UAT): Business/QA teams manually verify that the new feature meets the original business requirements in a production-	Business Sign-off



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Phase	Steps (A-Z Flow)	Key Goals
	like setting.	
	L. Performance & Security Testing: Load/stress tests and advanced Dynamic Application Security Testing (DAST) are run against the staging environment.	Scalability & Hardening
IV. Readiness & Approval	M. Change Management: A formal Request for Change (RFC) is submitted with all test reports (E-L) attached.	Governance
	N. Final Backup: Database and configuration backups of the production site are taken immediately before the deployment window starts.	Disaster Preparation
	O. Communication: Stakeholders (Support, Business) are notified of the upcoming "Go-Live."	Transparency
V. Production Deployment (Go-Live)	P. Environment Check: Final automated checks ensure production servers are healthy and ready to receive the update.	Pre-Flight Check
	Q. Deployment Execution: The deployment script (e.g., Blue/Green, Canary, Rolling Update) is executed automatically. This deploys the artifact (G) to the main site.	Code Swap
	R. Smoke Testing (Post-Deployment): A rapid set of essential, high-level tests are run <i>on the live site</i> to ensure core functions (login, checkout, homepage) are working immediately after the deployment.	Live Verification
VI. Monitoring & Feedback	S. Live Monitoring: Operations/DevOps teams actively monitor key metrics (error rates, CPU usage, latency) in real-time.	Health Check
	T. User Feedback: Feedback channels are monitored for immediate reports of issues.	Real-World Impact



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Phase	Steps (A-Z Flow)	Key Goals
	U. Triage and Fix: If a critical bug is found, the team either quickly implements a hotfix or executes the Rollback procedure (V).	Incident Response
	V. Rollback (If needed): The previous, stable version (N) is immediately deployed, reverting the system to the pre-deployment state.	Risk Mitigation
VII. Post-Mortem & Improvement	W. Post-Deployment Review (PDR): A meeting is held to review the deployment process, documenting any issues and calculating key metrics (e.g., Deployment Time, Rollback Rate).	Process Learning
	X. Clean-up: Old artifacts and temporary environments are decommissioned.	Cost Optimization
	Y. Documentation Update: The Runbook and internal knowledge base are updated with any lessons learned or new configuration details.	Knowledge Transfer
	Z. Iteration: The lessons learned (W) are fed back into the Planning phase of the next feature (looping back to Step A).	Continuous Improvement

Reminder:

Check

1. Header and footer before deploy
2. Clean unnecessary codes and or messages like .php etc
3. Check and check and check your work stability
4. Always contact your team leader at any time.
5. Be sure you always learn something new

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